

Shiv Nadar University Chennai

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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Experiment 3

Implementation of Convolutional Neural Networks (CNNs) for Image Classification

1. Objective

To understand the working principle of Convolutional Neural Networks by implementing convolution, pooling, feature map visualization, and image classification using TensorFlow/Keras.

2. Background Theory

A Convolutional Neural Network consists of

- Convolution Layer
- Activation Layer
- Pooling Layer
- Fully Connected Layer

The convolution operation is

$$Y(i, j) = \sum_m \sum_n X(i+m, j+n) K(m, n)$$

where

- X : Input Image
- K : Kernel (Filter)
- Y : Feature Map

The output feature map size is

$$\frac{N - F + 2P}{S} + 1$$

where

- N : Input Size
- F : Filter Size
- P : Padding
- S : Stride

3. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Classes : 10
- Image Size : $32 \times 32 \times 3$

4. Experimental Procedure

4.1 Dataset Exploration

Dataset Dimensions

Dataset	Dimensions
Training Images	(50000, 32, 32, 3)
Training Labels	(50000, 1)
Testing Images	(10000, 32, 32, 3)
Testing Labels	(10000, 1)

Sample Images



Figure 1: Sample Images from CIFAR-10

Observation:

These images are from different types of the CIFAR-10 dataset that consist of animals, automobiles, and many other things. Since the size of these images is very small (32×32), some parts of the images are hard to recognize.

Class Distribution

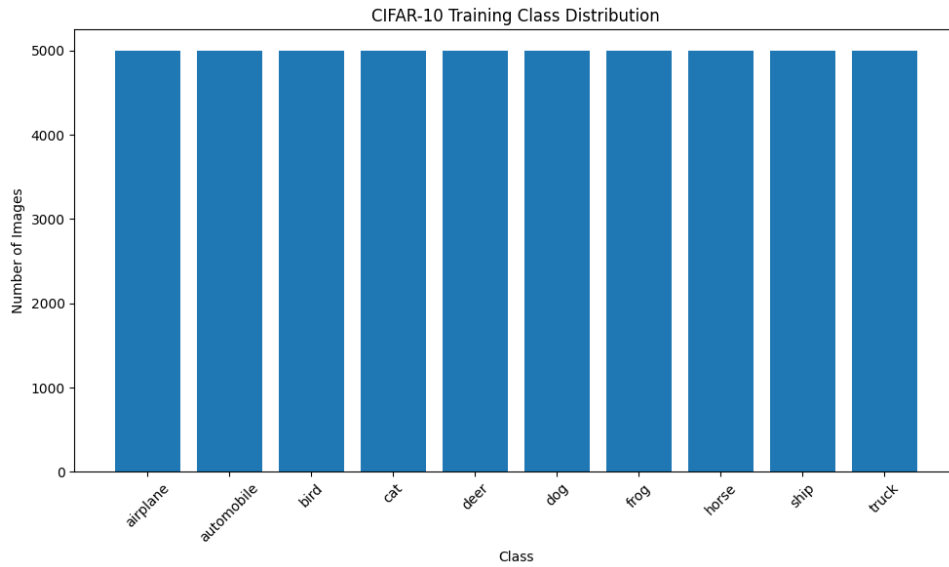


Figure 2: Class Distribution

Observation:

Class distribution is balanced as there are approximately 5,000 images for each of the 10 classes of CIFAR-10 available for training purposes. It is essential to have the balanced class distribution in order to prevent bias towards any particular class.

4.2 Data Preprocessing

The pixel values of the CIFAR-10 images, originally in the range $[0, 255]$, were normalized to the range $[0, 1]$ by dividing each pixel value by 255. This scaling improves numerical stability and speeds up convergence during training.

Stage	Data Type	Pixel Range
Before Normalization	uint8	$[0, 255]$
After Normalization	float32	$[0, 1]$

Observation:

The images in the CIFAR-10 dataset are made up of $32 \times 32 \times 3$ images in the RGB form, and they are inputted to the neural network in the correct way. This is made sure by pre-processing of the images along with their labels.

4.3 Model Construction

Construct the following CNN.

Input → Conv → ReLU → MaxPool → Conv → ReLU → MaxPool → Flatten → Dense → Softmax

Layer (Type)	Output Shape	Parameters
Conv2D	(None, 32, 32, 16)	448
ReLU	(None, 32, 32, 16)	0
MaxPooling2D	(None, 16, 16, 16)	0
Conv2D	(None, 16, 16, 32)	4,640
ReLU	(None, 16, 16, 32)	0
MaxPooling2D	(None, 8, 8, 32)	0
Flatten	(None, 2048)	0
Dense	(None, 128)	262,272
ReLU	(None, 128)	0
Dropout (0.4)	(None, 128)	0
Dense (Softmax)	(None, 10)	1,290
Total Trainable Parameters		268,650

Observation:

The CNN consists of convolution and ReLU layers which are followed by max pooling layers, flattening layer and fully connected layers for classification. The CNN is able to learn spatial features from the images and then classify the image based on these features.

4.4 Model Training

Train for

- Optimizer : Adam
- Epochs : 20
- Batch Size : 32

Training Accuracy

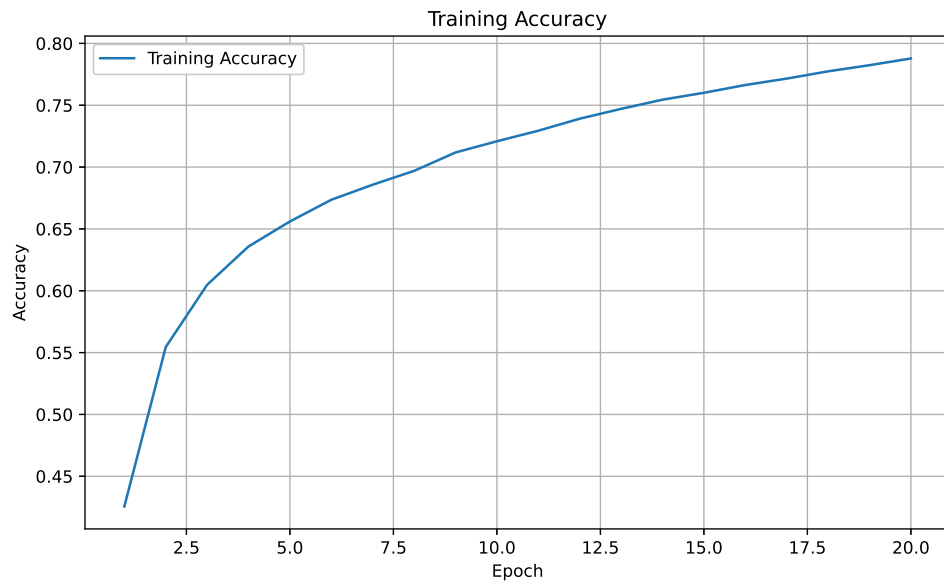


Figure 3: Training Accuracy

Observation:

The rate of training accuracy increases from approximately 42.6% to 78.8% between epochs 1 and 20. The increase in training accuracy signifies the ability of the neural network to learn more and more features from the training data.

Validation Accuracy

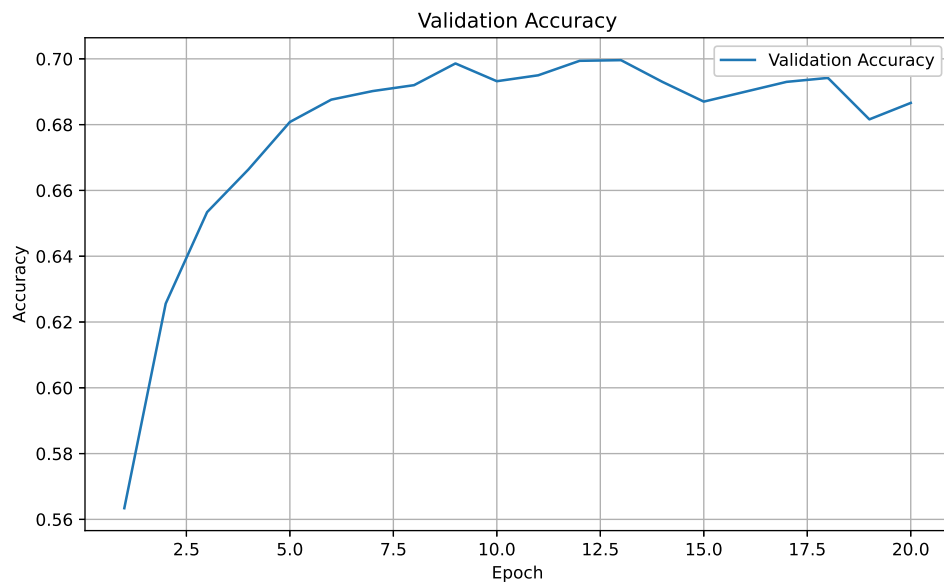


Figure 4: Validation Accuracy

Observation:

However, the accuracy level of the model spikes dramatically at the beginning and achieves its peak of about 70% in the thirteenth epoch. After this point, the graph starts to move erratically and decline, showing that the model becomes overfitted.

Training Loss

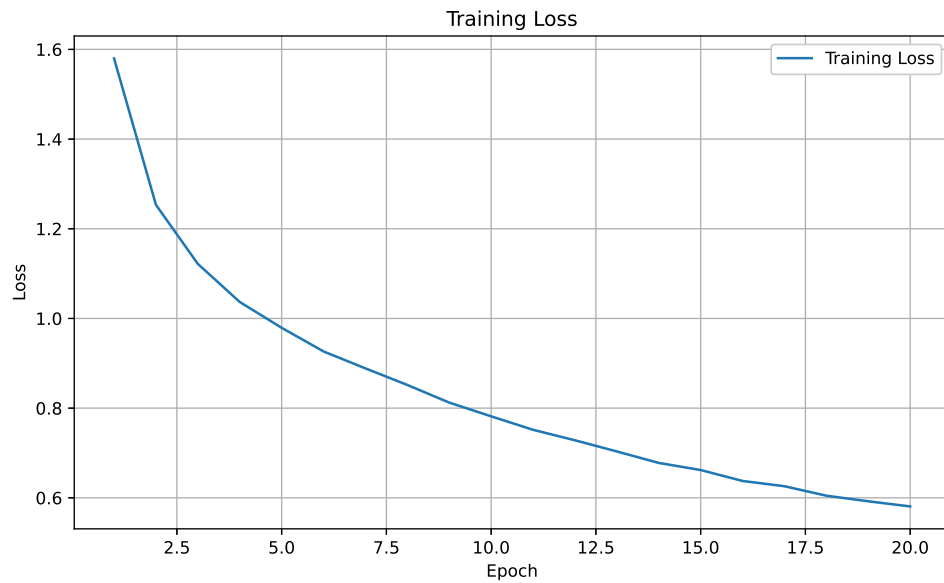


Figure 5: Training Loss

Observation:

The value of the training loss continues to drop during the 20 training epochs from about 1.58 to 0.58. It indicates that the performance of the model improves during the entire training process.

Validation Loss

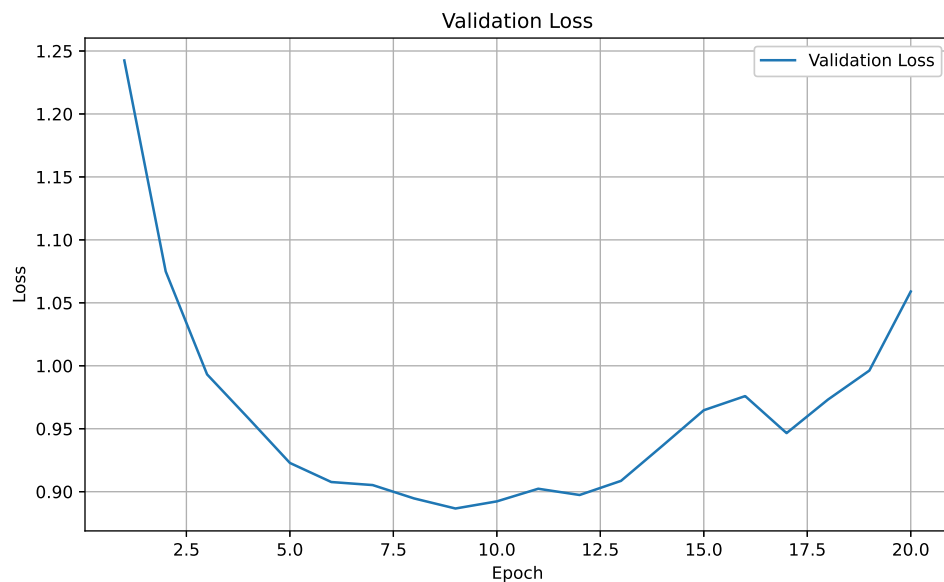


Figure 6: Validation Loss

Observation:

Validation loss begins to decrease, achieving its lowest point at approximately 0.887 at the ninth epoch. However, validation loss increases steadily after this, even as training loss continues to decrease. This means that overfitting is occurring during the later epochs.

Training Accuracy vs Validation Accuracy

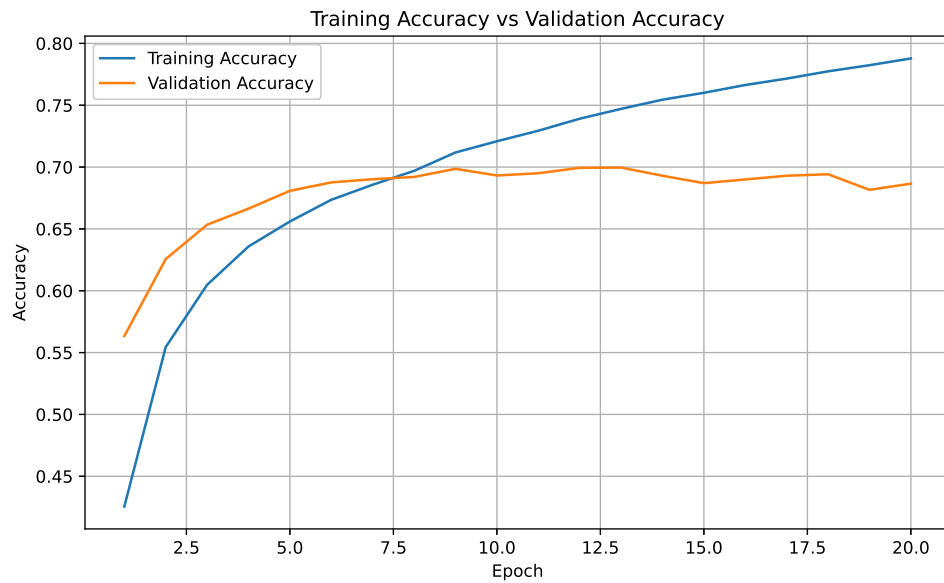


Figure 7: Training Accuracy vs Validation Accuracy

Observation:

For the first few epochs, there is a good alignment between the accuracy curves for training and validation, but as time passes, the training accuracy curve continues to go up steadily while the validation accuracy stays flat and oscillates in the range 68-70%.

Training Loss vs Validation Loss

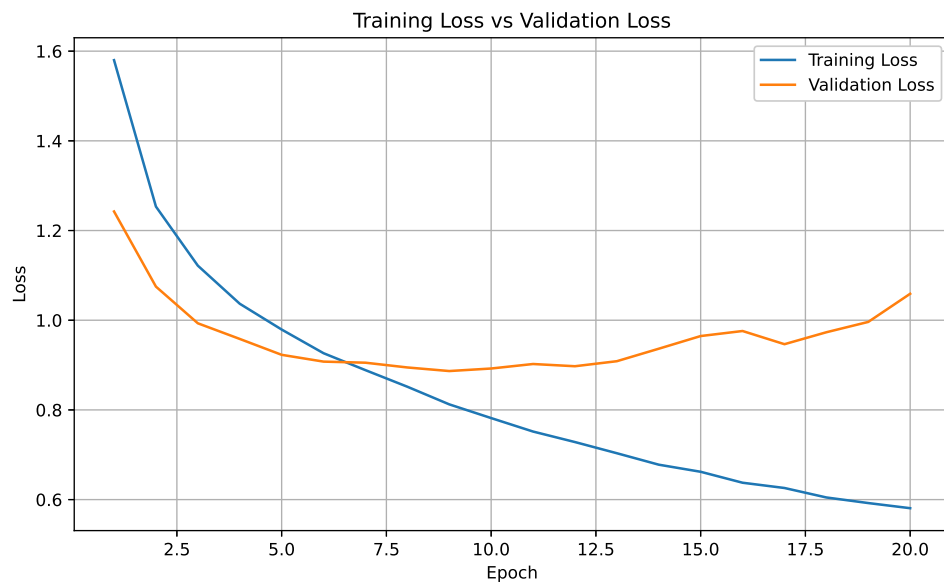


Figure 8: Training Loss vs Validation Loss

Observation:

Both training and validation losses fall simultaneously in the beginning epochs. Starting from epoch 9, however, the training loss decreases continuously while the validation loss increases, clearly showing that there is an overfitting problem in the latter stages of the training process.

4.5 Model Evaluation

Performance Metrics

Metric	Value
Accuracy	0.6751
Precision	0.6807
Recall	0.6751
F1-score	0.6745
Number of Parameters	268,650

Feature Maps

Display at least eight feature maps after the first convolution layer.

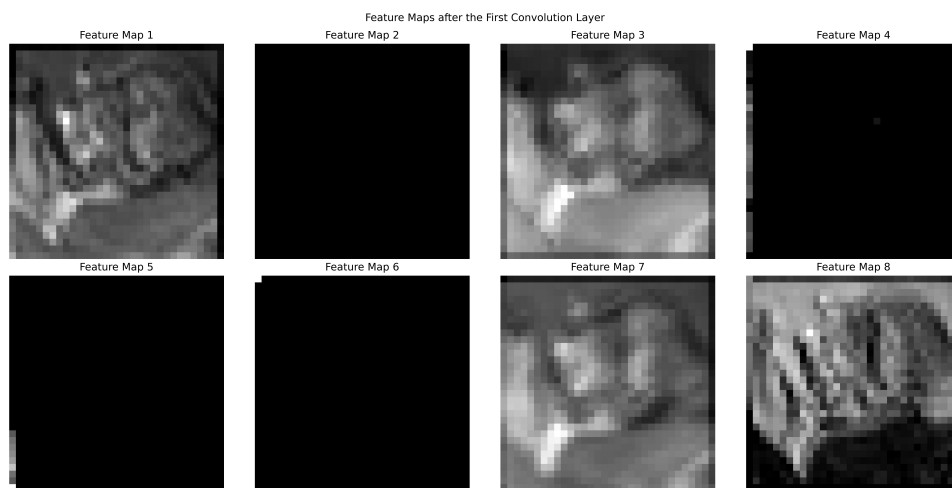


Figure 9: Feature Maps after the First Convolution Layer

Observation:

The feature maps created by the first convolutional layer have a preference for specific features such as edges, textures, contours, and contrast. The differences in the feature maps suggest that the filters detect certain aspects of the input image.

Confusion Matrix



Figure 10: Confusion Matrix

Observation:

From the confusion matrix, one can see that there are quite some good performances from the model in classifying images like the ship, truck, automobile, and airplane. But there have been more confusions with those objects that had similarity in their appearances such as cat and dog, bird and deer, bird and cat.

Classification Report

Class	Precision	Recall	F1-score	Support
airplane	0.73	0.68	0.70	1000
automobile	0.76	0.80	0.78	1000
bird	0.65	0.51	0.57	1000
cat	0.47	0.52	0.49	1000
deer	0.61	0.66	0.64	1000
dog	0.57	0.57	0.57	1000
frog	0.82	0.67	0.74	1000
horse	0.73	0.76	0.74	1000
ship	0.66	0.87	0.75	1000
truck	0.80	0.72	0.76	1000
Accuracy	0.68			10000
Macro Avg	0.68	0.68	0.67	10000
Weighted Avg	0.68	0.68	0.67	10000

Observation:

From the classification report results, it is evident that classes with unique appearances and colors, for example, car, ship, and truck, attain high values in terms of precision and recall, while classes like cat and dog with similar appearances obtain low values because of misclassifications.

5. CNN Experiments

5.1 Convolution Kernel Comparison

Implement a convolution layer and compare

- Kernel = 3×3
- Kernel = 5×5
- Kernel = 7×7

Record feature map sizes.

Kernel Size	Feature Map Size
3×3	(30, 30, 16)
5×5	(28, 28, 16)
7×7	(26, 26, 16)

Observation:

Increase in the size of the convolution kernel from 3×3 to 5×5 and 7×7 increases the receptive field of the convolution. Larger convolution kernels have a better capability of capturing large-scale spatial information, but the number of parameters increases.

5.2 Stride and Padding

Compare

- Stride = 1
- Stride = 2
- Padding = Same
- Padding = Valid

Computed output dimensions (using a 3×3 kernel with 16 filters on a $32 \times 32 \times 3$ input):

Stride	Padding	Output Dimension
1	Same	(32, 32, 16)
2	Same	(16, 16, 16)
1	Valid	(30, 30, 16)
2	Valid	(15, 15, 16)

Observation:

Through increasing the size of the stride, the spatial dimension of the outputted feature map becomes smaller, while the purpose of using padding is to keep the spatial information around the borders of the image. The ‘same’ padding normally preserves the size of the input data in case the size of stride equals to 1, otherwise ‘valid’ padding reduces the output size.

5.3 Feature Map Visualization

Visualize feature maps after the first convolution layer.
Display at least eight feature maps.

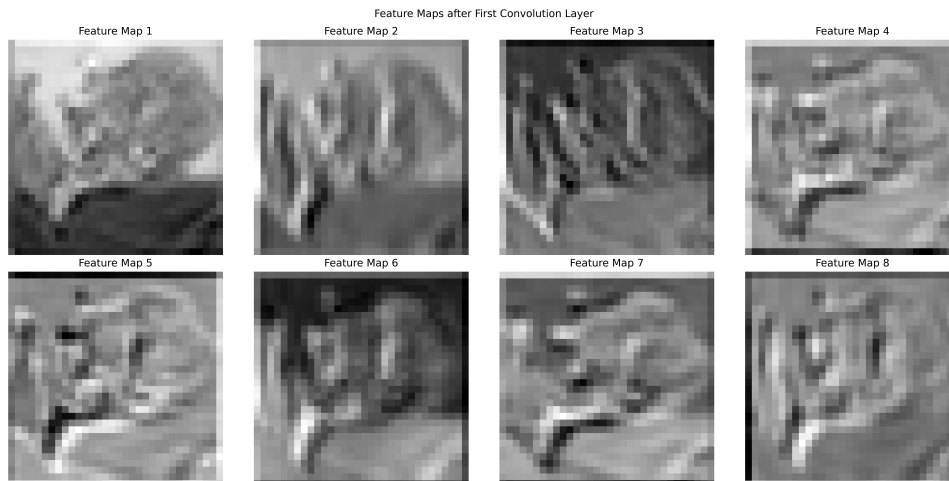


Figure 11: Feature Maps after the First Convolution Layer of the Trained CNN

Observation:

In the process of visualizing the model, it is clear that convolution layer number one learns features that are not found in the input image. Some of the feature maps detect edges and boundaries, while other feature maps will detect regions in the image.

5.4 Pooling Comparison

Compared:

- Max Pooling
- Average Pooling

Compared output size and accuracy:

Pooling Method	Output Size	Accuracy
Max Pooling	$8 \times 8 \times 32$	0.6751
Average Pooling	$8 \times 8 \times 32$	0.6889

Observation:

In Max Pooling, only the maximum activation value from the pooling region is retained, thereby helping to retain the feature while at the same time reducing its dimensionality. Conversely, in Average Pooling, the average activation value is computed and a smooth output is generated. In this case study, Average Pooling yielded a higher test accuracy (0.6889) compared to Max Pooling (0.6751).

6. Additional Exercises

1. Calculate the output size for a 64×64 image using a 5×5 kernel with stride 2 and padding 2.

Parameter	Value
Input Size (N)	64
Kernel Size (F)	5
Stride (S)	2
Padding (P)	2
Output Size	32×32

$$\text{Output} = \left\lfloor \frac{64 - 5 + 2(2)}{2} \right\rfloor + 1 = 32$$

2. Compute the trainable parameters of a convolution layer having 64 filters of size 3×3 and an RGB input.

Parameter	Value
Filters	64
Kernel Size	3×3
Input Channels	3
Trainable Parameters	$(3 \times 3 \times 3 + 1) \times 64 = 1,792$

3. Compare ReLU and Sigmoid activation functions.

Property	ReLU	Sigmoid
Function	$\max(0, x)$	$1/(1 + e^{-x})$
Output Range	$[0, \infty)$	$(0, 1)$
Common Use	Hidden layers	Binary output
Vanishing Gradient	Lower risk	Higher risk
Computation	Simple	More expensive

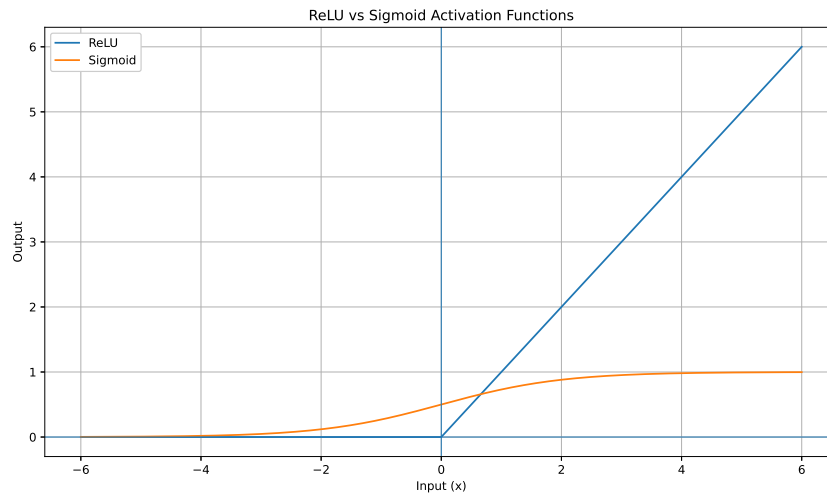


Figure 12: ReLU vs Sigmoid Activation Functions

Observation:

The output for ReLU is zero for negative values and increases linearly with positive values. The sigmoid function has an output in the range of 0 and 1 but it tends to saturate with increasing positive and negative values.

4. Replace Max Pooling with Average Pooling and compare the results.

Metric	Value
Max Pooling Test Accuracy	0.6751
Average Pooling Test Accuracy	0.6889
Accuracy Difference (Max – Average)	–0.0138
Better Performing Method	Average Pooling

Observation:

The average pooling resulted in more accurate test results compared to max pooling in this particular study, which means that the smoothing out of activations is somewhat better than picking up the maximum value.

5. Increase the number of convolution filters from 16 to 64 and analyze the effect on accuracy and computation time.

Model	Parameters	Test Accuracy	Time (sec)
16 Filters	268,650	0.6264	110.67
64 Filters	1,125,642	0.6698	138.44

Observation:

By increasing the number of filters from 16 to 64, there are more features to be learned by the CNN, resulting in an increase in test accuracy from 0.6264 to 0.6698. On the other hand, the new model has more trainable parameters and has consumed more processing time (138.44s vs. 110.67s).

7. Results

Record

- Training Accuracy
- Testing Accuracy
- Precision
- Recall
- F1-score
- Number of Parameters

Metric	Value
Training Accuracy	0.7878
Testing Accuracy	0.6751
Precision	0.6807
Recall	0.6751
F1-score	0.6745
Number of Parameters	268,650

8. Discussion

Answer the following.

1. Why is convolution preferred over fully connected layers for images?

Convolution is better as it retains the image structure and helps in identifying local features like edges and texture. Moreover, convolution uses weight sharing, and therefore, it requires fewer parameters than fully connected layers.

2. How does stride affect the feature map size?

The stride is responsible for the distance that the convolution filter travels over the image. The higher the stride, the lower the feature map will be; whereas, a lower stride will yield a higher feature map.

3. What is the role of padding?

Padding provides additional pixels, which are generally located around the edges of the input image. It helps in retaining the dimensions of the space and ensures that convolution can use data from the vicinity of the edges of the image.

4. Why is pooling used?

Pooling compresses the spatial dimensions of the feature map but still preserves the critical information. It also reduces the computational burden and may allow the network to be insensitive to small displacements of features.

5. How do feature maps represent image characteristics?

Feature maps depict the output generated from applying convolutional filters while recognizing specific patterns within the image. The filters can recognize patterns including edges, texture, boundaries, and other visual elements.

6. Why do CNNs require fewer parameters than MLPs?

The number of parameters needed by CNNs is much smaller due to the use of local connections and weight sharing. Local connections mean that the same filter is used on the various parts of the image rather than using distinct weights for each input pixel.

9. Conclusion

The experiment described the application of the Convolutional Neural Network approach for the problem of image classification on the CIFAR-10 dataset. Numerical examples demonstrated the process of convolution, max pooling, and calculation of trainable parameters, while CNN experiments helped reveal the role of kernel size, stride, padding, and pooling in the process of feature extraction.

From the training results, it is evident that the model has been trained well, as training accuracy improved up to 78.8% and training loss decreased significantly throughout 20 epochs. At the same time, validation accuracy peaked at around 70% and then began to fluctuate and decline, while validation loss reached its minimum around the ninth epoch before rising steadily. This means that the model faced the issue of overfitting in the later epochs.

The feature map plots showed how the convolutional filters capture various low-level features like edges, texture, and boundaries. The confusion matrix additionally demonstrated how the model has correctly classified most of the categories but had trouble with similar images, such as cats and dogs.

10. References

1. Goodfellow et al., *Deep Learning*.
2. Bishop, *Pattern Recognition and Machine Learning*.
3. Haykin, *Neural Networks and Learning Machines*.
4. TensorFlow Documentation.
5. CIFAR-10 Dataset Documentation.